Activity-Based Risk Assessment - Structural Work Pack

Activity-Based Risk Assessment Matrix (5-5 Initial - Residual)

Work Package: Structural Foundation & Steel Erection - Matrix: 5-5 Likelihood - Severity

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01. Sub-structure Excavation
Hazard & Consequence: Underground live cable strike / trench collapse
Init Risk: 20 (High)
Controls & Hold Points: PTW + CAT scan + trial pit hand-dig [HOLD POINT]
Res: 4 (Low)

02. Mobile Crane Rigging
Hazard & Consequence: Load drop / outrigger ground bearing failure
Init Risk: 16 (High)
Controls & Hold Points: 3rd-party lift plan, spreader mats, anemometer <20 kt
Res: 4 (Low)

03. Steel Column Erection
Hazard & Consequence: Fall from height / Pinch point during bolt-up
Init Risk: 15 (High)
Controls & Hold Points: MEWP rescue plan, 100% twin-lanyard tie-off, taglines
Res: 3 (Low)

04. Grouting & Chemical Curing
Hazard & Consequence: Dermal/respiratory exposure to epoxy grouts
Init Risk: 10 (Med)
Controls & Hold Points: COSHH assessment + SDS on site + nitrile/respirator PPE
Res: 2 (Low)

5-5 Risk Evaluation Key & Client HSE Authority Sign-Off

Residual risk scores above 8 (Medium) require formal client/site HSE authority review before PTW issuance.

1-3. Elimination & Engineering
Verification Method: Temporary shoring design & certified lifting gear register
Owner Role: Package Engineer
Gate Status: [HOLD POINT 01]
Mandatory

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4. Administrative & PTW
Verification Method: Pre-shift TBT sign-off & exclusion zone barricading
Owner Role: Site HSE Officer
Gate Status: Shift Inspection
Active

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5. PPE & Emergency Rescue
Verification Method: Full body harness EN 361 + tripod retrieval standby
Owner Role: Rescue Team Lead
Gate Status: Verified Pre-Start
Checked

Activity Matrix - Earthworks & Substructure

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Earthworks - Bulk Excavation
Hazard & Consequence: Plant-pedestrian interface / reversing plant strike
Init Risk: 20 (High)
Controls & Hold Points: Trained banksman, segregated walkways, reversing cameras [HOLD POINT: daily plant check]
Res: 5 (Low)

Earthworks - Trench Support
Hazard & Consequence: Trench wall collapse / engulfment
Init Risk: 25 (High)
Controls & Hold Points: Batterered or shored sides per soil class, ladder access every 15 m, daily competent-person inspection [HOLD POINT]
Res: 5 (Low)

Formwork & Falsework Erection
Hazard & Consequence: Falsework overload / premature striking
Init Risk: 20 (High)
Controls & Hold Points: Designed falsework with tag system, striking only on engineer permit-to-strike [HOLD POINT]
Res: 4 (Low)

Reinforcement Fixing
Hazard & Consequence: Manual handling strain / protruding bar impalement
Init Risk: 12 (Med)
Controls & Hold Points: Mechanical lifting for bundles, mushroom caps on starter bars, team-lift limits
Res: 4 (Low)

Activity-Based Risk Assessment - Structural Work Pack

Activity Matrix - Formwork & Rebar

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Concreting - Pump & Skip
Hazard & Consequence: Hose whip / concrete splash to eyes
Init Risk: 15 (High)
Controls & Hold Points: Secured hose couplings with whip checks, sealed goggles, exclusion under skip path
Res: 5 (Low)

Blockwork & Plastering
Hazard & Consequence: Falling blocks / silica dust inhalation
Init Risk: 12 (Med)
Controls & Hold Points: Toe boards on scaffolds, wet cutting + FFP3 respirators, dust extraction
Res: 4 (Low)

Structural Steel - Bolt-up
Hazard & Consequence: Dropped tools / falling objects
Init Risk: 20 (High)
Controls & Hold Points: Tool lanyards, bolt bags, hard-barricaded drop zone below [HOLD POINT: wind limit 20 kt]
Res: 5 (Low)

Roofing Works
Hazard & Consequence: Fall through fragile surface
Init Risk: 25 (High)
Controls & Hold Points: Crawling boards, edge protection, harness anchor plan [HOLD POINT: fragile surface permit]
Res: 5 (Low)

Activity Matrix - Steel & Roofing

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

MEP First Fix - Containment
Hazard & Consequence: Work at height from podium steps / cable drum handling
Init Risk: 12 (Med)
Controls & Hold Points: Podium steps with gate closed, drum jacks and chocks, two-person drum moves
Res: 3 (Low)

MEP Second Fix - Terminations
Hazard & Consequence: Electric shock / arc flash during termination
Init Risk: 20 (High)
Controls & Hold Points: LOTO applied and proven dead, insulated tools, competent electrician only [HOLD POINT]
Res: 4 (Low)

Finishing - Painting & Coating
Hazard & Consequence: VOC exposure / flammable vapour ignition
Init Risk: 15 (High)
Controls & Hold Points: Ventilation, COSHH assessment at point of use, no hot works in zone, respirators A2P3
Res: 5 (Low)

External Works - Kerbs & Paving
Hazard & Consequence: Disc cutter kickback / HAVS
Init Risk: 15 (High)
Controls & Hold Points: Guard-fitted cutters, HAVS exposure log, water suppression for dust
Res: 4 (Low)

Activity Matrix - MEP & Finishing

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Commissioning - Pressure Testing
Hazard & Consequence: Stored energy release / hose failure
Init Risk: 20 (High)
Controls & Hold Points: Test pack approved, barricaded test zone, calibrated gauges, gradual pressurisation [HOLD POINT]
Res: 5 (Low)

Handover - Snagging Works
Hazard & Consequence: Lone working / incomplete isolations
Init Risk: 10 (Med)
Controls & Hold Points: Permit-controlled snagging list, buddy system, isolation register check
Res: 3 (Low)

Material Deliveries & Offloading
Hazard & Consequence: Load shift / forklift pedestrian strike
Init Risk: 16 (High)
Controls & Hold Points: Delivery management plan, trained slinger/signaller, exclusion during offload
Res: 4 (Low)

Waste Handling & Housekeeping
Hazard & Consequence: Slips and trips / sharps injury
Init Risk: 9 (Med)
Controls & Hold Points: Segregated waste streams, daily housekeeping inspection, sharps gloves for sorters
Res: 3 (Low)

Activity Matrix - Temporary Works & Electrics

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Site Access & Security
Hazard & Consequence: Unauthorised entry / public interface
Init Risk: 12 (Med)
Controls & Hold Points: Turnstile or gate log, visitor induction, hoarding integrity checks
Res: 3 (Low)

Temporary Electrics - Distribution
Hazard & Consequence: Damaged leads / RCD failure
Init Risk: 15 (High)
Controls & Hold Points: 110V or RCD-protected supplies, weekly PAT/inspection register, cable ramps
Res: 4 (Low)

Hot Works - Welding & Cutting
Hazard & Consequence: Fire / sparks to combustibles
Init Risk: 20 (High)
Controls & Hold Points: Hot work permit, fire watch during + 60 min after, extinguishers at point of work [HOLD POINT]
Res: 5 (Low)

Work at Height - Scaffold Use
Hazard & Consequence: Fall of person / incomplete scaffold
Init Risk: 25 (High)
Controls & Hold Points: Scafftag system, weekly + after-alteration inspections, no unauthorised alterations [HOLD POINT]
Res: 5 (Low)

Activity Matrix - Hot Works & Height

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Lifting - Mobile Crane General
Hazard & Consequence: Overload / load swing
Init Risk: 20 (High)
Controls & Hold Points: Lift plan per lift category, certified gear register, anemometer monitoring [HOLD POINT]
Res: 4 (Low)

Plant Refuelling
Hazard & Consequence: Fuel spill / fire
Init Risk: 12 (Med)
Controls & Hold Points: Bunded bowser, drip trays, spill kit at point of use, no smoking zone
Res: 3 (Low)

Night Works
Hazard & Consequence: Poor visibility / fatigue
Init Risk: 15 (High)
Controls & Hold Points: Lighting survey (min lux at task), fatigue roster limits, hi-vis Class 3
Res: 5 (Low)

Heat Stress Management
Hazard & Consequence: Heat exhaustion / heat stroke
Init Risk: 16 (High)
Controls & Hold Points: WBGT monitoring, shaded rest + cool water, acclimatisation schedule, buddy checks
Res: 4 (Low)

Activity Matrix - Plant & Logistics

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Dust-Generating Works
Hazard & Consequence: Respirable crystalline silica
Init Risk: 16 (High)
Controls & Hold Points: Water suppression, on-tool extraction, FFP3 face-fit tested, health surveillance
Res: 4 (Low)

Noise-Intensive Works
Hazard & Consequence: Noise-induced hearing loss
Init Risk: 12 (Med)
Controls & Hold Points: Noise zones signed, hearing protection mandatory >85 dB(A), plant selection low-noise first
Res: 3 (Low)

Traffic Management - Internal Roads
Hazard & Consequence: Vehicle collision at pinch points
Init Risk: 16 (High)
Controls & Hold Points: One-way system, 10 km/h limit, passing bays, trained traffic marshals
Res: 4 (Low)

Welfare - Water, Rest, Sanitation
Hazard & Consequence: Dehydration / hygiene illness
Init Risk: 9 (Med)
Controls & Hold Points: Potable water points, shaded rest areas, serviced sanitation per headcount ratio
Res: 2 (Low)

Activity Matrix - Health Hazards

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Adverse Weather - Lightning & Storm
Hazard & Consequence: Lightning strike / crane instability
Init Risk: 15 (High)
Controls & Hold Points: 30-30 lightning rule, crane stand-down limits, secure loose materials
Res: 5 (Low)

Housekeeping - Access Routes
Hazard & Consequence: Slips, trips and falls (same level)
Init Risk: 12 (Med)
Controls & Hold Points: Designated walkways kept clear, cable covers, daily supervisor walk-round
Res: 3 (Low)

Visitor & Delivery Driver Safety
Hazard & Consequence: Unfamiliar persons in live work zones
Init Risk: 12 (Med)
Controls & Hold Points: Escorted access only, driver briefing card, PPE issue at gate
Res: 3 (Low)

Document Control & Record Retention
Hazard & Consequence: Superseded revision used in field
Init Risk: 9 (Med)
Controls & Hold Points: Controlled copy register, revision check at TBT, obsolete copies withdrawn
Res: 2 (Low)

Activity Matrix - Traffic & Welfare

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Incident Reporting & Escalation
Hazard & Consequence: Delayed reporting / evidence loss
Init Risk: 12 (Med)
Controls & Hold Points: Immediate verbal + 24h written report, scene preservation, investigation per severity matrix
Res: 3 (Low)

Subcontractor Coordination (SIMOPS)
Hazard & Consequence: Conflicting simultaneous operations
Init Risk: 16 (High)
Controls & Hold Points: Daily SIMOPS meeting, shared permit board, interface risk review [HOLD POINT]
Res: 4 (Low)

Manual Handling - General
Hazard & Consequence: Musculoskeletal injury
Init Risk: 12 (Med)
Controls & Hold Points: Mechanical aids first, load weight labels, manual handling training records
Res: 3 (Low)

Storage of Flammables & Gas Cylinders
Hazard & Consequence: Fire / cylinder projectile
Init Risk: 16 (High)
Controls & Hold Points: Ventilated cage, upright chained cylinders, separation distances, signage
Res: 4 (Low)

Activity Matrix - Weather & Fatigue

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Excavation Services - Trial Pits
Hazard & Consequence: Unknown service strike
Init Risk: 20 (High)
Controls & Hold Points: Drawings + CAT scan + hand-dig only within 500 mm of marked services [HOLD POINT]
Res: 5 (Low)

Concrete Washout
Hazard & Consequence: Alkaline runoff to ground/drain
Init Risk: 9 (Med)
Controls & Hold Points: Designated lined washout pit only, pH check before disposal
Res: 2 (Low)

Tower Crane - Climbing Operations
Hazard & Consequence: Structural overload during climb
Init Risk: 25 (High)
Controls & Hold Points: OEM climbing procedure, specialist crew, wind hold limits [HOLD POINT: climb permit]
Res: 5 (Low)

Demolition - Soft Strip
Hazard & Consequence: Hidden asbestos / structural instability
Init Risk: 20 (High)
Controls & Hold Points: Pre-demolition survey, licensed removal where identified, engineer sequence sign-off [HOLD POINT]
Res: 5 (Low)

Activity Matrix - Interfaces & Public

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Piling - Bored Operations
Hazard & Consequence: Auger entanglement / bore collapse
Init Risk: 20 (High)
Controls & Hold Points: Auger guards, exclusion zone, bore capping same shift
Res: 5 (Low)

Roadworks Interface - Public Highway
Hazard & Consequence: Public vehicle incursion
Init Risk: 20 (High)
Controls & Hold Points: Approved traffic management layout, advance signage, barrier-rated separation [HOLD POINT]
Res: 5 (Low)

Confined Welfare - Drinking Water Quality
Hazard & Consequence: Contaminated supply illness
Init Risk: 9 (Med)
Controls & Hold Points: Tested potable supply, dated containers, cleaning schedule
Res: 2 (Low)

Fatigue - Extended Shifts
Hazard & Consequence: Error from exhaustion on safety-critical tasks
Init Risk: 15 (High)
Controls & Hold Points: Max shift + rest rules, no safety-critical solo work after extended hours
Res: 4 (Low)

Activity Matrix - Specialist Operations

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Young & Inexperienced Workers
Hazard & Consequence: Unrecognised hazards
Init Risk: 12 (Med)
Controls & Hold Points: Named supervisor, restricted task list, enhanced supervision ratios
Res: 3 (Low)

Emergency Drill Programme
Hazard & Consequence: Unpractised evacuation / slow muster
Init Risk: 12 (Med)
Controls & Hold Points: Monthly drill rotation (fire, rescue, spill), timed muster with roll call
Res: 3 (Low)

First Aid Provision
Hazard & Consequence: Inadequate response to injury
Init Risk: 12 (Med)
Controls & Hold Points: Trained first aiders per shift ratio, stocked stations, hospital route posted
Res: 3 (Low)

PPE Issue & Compliance
Hazard & Consequence: Non-use / damaged PPE
Init Risk: 9 (Med)
Controls & Hold Points: Free issue + fit checks, PPE matrix by task, supervisor spot checks
Res: 2 (Low)

Activity Matrix - Management Controls

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Safety Signage & Barriers Audit
Hazard & Consequence: Missing or ignored warnings
Init Risk: 9 (Med)
Controls & Hold Points: Weekly audit walk, multilingual pictorial signage, barrier integrity checks
Res: 2 (Low)

Toolbox Talk Programme
Hazard & Consequence: Generic talks with no task relevance
Init Risk: 9 (Med)
Controls & Hold Points: Task-specific TBT before each shift, attendance signed, topics logged weekly
Res: 2 (Low)

Management Site Tours
Hazard & Consequence: Hazards normalised at worker level
Init Risk: 9 (Med)
Controls & Hold Points: Weekly leadership tour with close-out actions tracked to completion
Res: 2 (Low)

Change Management - Scope Variation
Hazard & Consequence: Unassessed new hazards from changed method
Init Risk: 16 (High)
Controls & Hold Points: MOC form before variation starts, RA revision re-briefed to crew [HOLD POINT]
Res: 4 (Low)

Activity Matrix - Close-Out

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Close-Out - Demobilisation
Hazard & Consequence: Residual hazards left for others
Init Risk: 10 (Med)
Controls & Hold Points: Demob checklist, services capped and marked, final joint inspection
Res: 3 (Low)

Lessons Learned Register
Hazard & Consequence: Repeat incidents across packages
Init Risk: 9 (Med)
Controls & Hold Points: Register reviewed monthly, actions briefed at TBT, trends to management review
Res: 2 (Low)

Audit & Inspection Schedule
Hazard & Consequence: Drift from planned controls
Init Risk: 9 (Med)
Controls & Hold Points: Weekly HSE inspection, monthly audit, findings tracked with owners and dates
Res: 2 (Low)

Permit Office - Daily Board Control
Hazard & Consequence: Conflicting live permits
Init Risk: 16 (High)
Controls & Hold Points: Permit board reviewed at shift start, conflicts resolved before issue [HOLD POINT]
Res: 4 (Low)

Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Site Access & Security
Hazard & Consequence: Unauthorised entry / public interface
Init Risk: 12 (Med)
Controls & Hold Points: Turnstile or gate log, visitor induction, hoarding integrity checks
Res: 3 (Low)

Temporary Electrics - Distribution
Hazard & Consequence: Damaged leads / RCD failure
Init Risk: 15 (High)
Controls & Hold Points: 110V or RCD-protected supplies, weekly PAT/inspection register, cable ramps
Res: 4 (Low)

Hot Works - Welding & Cutting
Hazard & Consequence: Fire / sparks to combustibles
Init Risk: 20 (High)
Controls & Hold Points: Hot work permit, fire watch during + 60 min after, extinguishers at point of work [HOLD POINT]
Res: 5 (Low)

Work at Height - Scaffold Use
Hazard & Consequence: Fall of person / incomplete scaffold
Init Risk: 25 (High)
Controls & Hold Points: Scafftag system, weekly + after-alteration inspections, no unauthorised alterations [HOLD POINT]
Res: 5 (Low)

Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Lifting - Mobile Crane General
Hazard & Consequence: Overload / load swing
Init Risk: 20 (High)
Controls & Hold Points: Lift plan per lift category, certified gear register, anemometer monitoring [HOLD POINT]
Res: 4 (Low)

Plant Refuelling
Hazard & Consequence: Fuel spill / fire
Init Risk: 12 (Med)
Controls & Hold Points: Bunded bowser, drip trays, spill kit at point of use, no smoking zone
Res: 3 (Low)

Night Works
Hazard & Consequence: Poor visibility / fatigue
Init Risk: 15 (High)
Controls & Hold Points: Lighting survey (min lux at task), fatigue roster limits, hi-vis Class 3
Res: 5 (Low)

Heat Stress Management
Hazard & Consequence: Heat exhaustion / heat stroke
Init Risk: 16 (High)
Controls & Hold Points: WBGT monitoring, shaded rest + cool water, acclimatisation schedule, buddy checks
Res: 4 (Low)

Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Dust-Generating Works
Hazard & Consequence: Respirable crystalline silica
Init Risk: 16 (High)
Controls & Hold Points: Water suppression, on-tool extraction, FFP3 face-fit tested, health surveillance
Res: 4 (Low)

Noise-Intensive Works
Hazard & Consequence: Noise-induced hearing loss
Init Risk: 12 (Med)
Controls & Hold Points: Noise zones signed, hearing protection mandatory >85 dB(A), plant selection low-noise first
Res: 3 (Low)

Traffic Management - Internal Roads
Hazard & Consequence: Vehicle collision at pinch points
Init Risk: 16 (High)
Controls & Hold Points: One-way system, 10 km/h limit, passing bays, trained traffic marshals
Res: 4 (Low)

Welfare - Water, Rest, Sanitation
Hazard & Consequence: Dehydration / hygiene illness
Init Risk: 9 (Med)
Controls & Hold Points: Potable water points, shaded rest areas, serviced sanitation per headcount ratio
Res: 2 (Low)

Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Adverse Weather - Lightning & Storm
Hazard & Consequence: Lightning strike / crane instability
Init Risk: 15 (High)
Controls & Hold Points: 30-30 lightning rule, crane stand-down limits, secure loose materials
Res: 5 (Low)

Housekeeping - Access Routes
Hazard & Consequence: Slips, trips and falls (same level)
Init Risk: 12 (Med)
Controls & Hold Points: Designated walkways kept clear, cable covers, daily supervisor walk-round
Res: 3 (Low)

Visitor & Delivery Driver Safety
Hazard & Consequence: Unfamiliar persons in live work zones
Init Risk: 12 (Med)
Controls & Hold Points: Escorted access only, driver briefing card, PPE issue at gate
Res: 3 (Low)

Document Control & Record Retention
Hazard & Consequence: Superseded revision used in field
Init Risk: 9 (Med)
Controls & Hold Points: Controlled copy register, revision check at TBT, obsolete copies withdrawn
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Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Incident Reporting & Escalation
Hazard & Consequence: Delayed reporting / evidence loss
Init Risk: 12 (Med)
Controls & Hold Points: Immediate verbal + 24h written report, scene preservation, investigation per severity matrix
Res: 3 (Low)

Subcontractor Coordination (SIMOPS)
Hazard & Consequence: Conflicting simultaneous operations
Init Risk: 16 (High)
Controls & Hold Points: Daily SIMOPS meeting, shared permit board, interface risk review [HOLD POINT]
Res: 4 (Low)

Manual Handling - General
Hazard & Consequence: Musculoskeletal injury
Init Risk: 12 (Med)
Controls & Hold Points: Mechanical aids first, load weight labels, manual handling training records
Res: 3 (Low)

Storage of Flammables & Gas Cylinders
Hazard & Consequence: Fire / cylinder projectile
Init Risk: 16 (High)
Controls & Hold Points: Ventilated cage, upright chained cylinders, separation distances, signage
Res: 4 (Low)

Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Excavation Services - Trial Pits
Hazard & Consequence: Unknown service strike
Init Risk: 20 (High)
Controls & Hold Points: Drawings + CAT scan + hand-dig only within 500 mm of marked services [HOLD POINT]
Res: 5 (Low)

Concrete Washout
Hazard & Consequence: Alkaline runoff to ground/drain
Init Risk: 9 (Med)
Controls & Hold Points: Designated lined washout pit only, pH check before disposal
Res: 2 (Low)

Tower Crane - Climbing Operations
Hazard & Consequence: Structural overload during climb
Init Risk: 25 (High)
Controls & Hold Points: OEM climbing procedure, specialist crew, wind hold limits [HOLD POINT: climb permit]
Res: 5 (Low)

Demolition - Soft Strip
Hazard & Consequence: Hidden asbestos / structural instability
Init Risk: 20 (High)
Controls & Hold Points: Pre-demolition survey, licensed removal where identified, engineer sequence sign-off [HOLD POINT]
Res: 5 (Low)

Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Piling - Bored Operations
Hazard & Consequence: Auger entanglement / bore collapse
Init Risk: 20 (High)
Controls & Hold Points: Auger guards, exclusion zone, bore capping same shift
Res: 5 (Low)

Roadworks Interface - Public Highway
Hazard & Consequence: Public vehicle incursion
Init Risk: 20 (High)
Controls & Hold Points: Approved traffic management layout, advance signage, barrier-rated separation [HOLD POINT]
Res: 5 (Low)

Confined Welfare - Drinking Water Quality
Hazard & Consequence: Contaminated supply illness
Init Risk: 9 (Med)
Controls & Hold Points: Tested potable supply, dated containers, cleaning schedule
Res: 2 (Low)

Fatigue - Extended Shifts
Hazard & Consequence: Error from exhaustion on safety-critical tasks
Init Risk: 15 (High)
Controls & Hold Points: Max shift + rest rules, no safety-critical solo work after extended hours
Res: 4 (Low)

Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Young & Inexperienced Workers
Hazard & Consequence: Unrecognised hazards
Init Risk: 12 (Med)
Controls & Hold Points: Named supervisor, restricted task list, enhanced supervision ratios
Res: 3 (Low)

Emergency Drill Programme
Hazard & Consequence: Unpractised evacuation / slow muster
Init Risk: 12 (Med)
Controls & Hold Points: Monthly drill rotation (fire, rescue, spill), timed muster with roll call
Res: 3 (Low)

First Aid Provision
Hazard & Consequence: Inadequate response to injury
Init Risk: 12 (Med)
Controls & Hold Points: Trained first aiders per shift ratio, stocked stations, hospital route posted
Res: 3 (Low)

PPE Issue & Compliance
Hazard & Consequence: Non-use / damaged PPE
Init Risk: 9 (Med)
Controls & Hold Points: Free issue + fit checks, PPE matrix by task, supervisor spot checks
Res: 2 (Low)

Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Safety Signage & Barriers Audit
Hazard & Consequence: Missing or ignored warnings
Init Risk: 9 (Med)
Controls & Hold Points: Weekly audit walk, multilingual pictorial signage, barrier integrity checks
Res: 2 (Low)

Toolbox Talk Programme
Hazard & Consequence: Generic talks with no task relevance
Init Risk: 9 (Med)
Controls & Hold Points: Task-specific TBT before each shift, attendance signed, topics logged weekly
Res: 2 (Low)

Management Site Tours
Hazard & Consequence: Hazards normalised at worker level
Init Risk: 9 (Med)
Controls & Hold Points: Weekly leadership tour with close-out actions tracked to completion
Res: 2 (Low)

Change Management - Scope Variation
Hazard & Consequence: Unassessed new hazards from changed method
Init Risk: 16 (High)
Controls & Hold Points: MOC form before variation starts, RA revision re-briefed to crew [HOLD POINT]
Res: 4 (Low)

Activity Matrix - Continuation Part 2

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Close-Out - Demobilisation
Hazard & Consequence: Residual hazards left for others
Init Risk: 10 (Med)
Controls & Hold Points: Demob checklist, services capped and marked, final joint inspection
Res: 3 (Low)

Lessons Learned Register
Hazard & Consequence: Repeat incidents across packages
Init Risk: 9 (Med)
Controls & Hold Points: Register reviewed monthly, actions briefed at TBT, trends to management review
Res: 2 (Low)

Audit & Inspection Schedule
Hazard & Consequence: Drift from planned controls
Init Risk: 9 (Med)
Controls & Hold Points: Weekly HSE inspection, monthly audit, findings tracked with owners and dates
Res: 2 (Low)

Permit Office - Daily Board Control
Hazard & Consequence: Conflicting live permits
Init Risk: 16 (High)
Controls & Hold Points: Permit board reviewed at shift start, conflicts resolved before issue [HOLD POINT]
Res: 4 (Low)

Activity Matrix - Continuation Part 3

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Lifting - Mobile Crane General
Hazard & Consequence: Overload / load swing
Init Risk: 20 (High)
Controls & Hold Points: Lift plan per lift category, certified gear register, anemometer monitoring [HOLD POINT]
Res: 4 (Low)

Plant Refuelling
Hazard & Consequence: Fuel spill / fire
Init Risk: 12 (Med)
Controls & Hold Points: Bunded bowser, drip trays, spill kit at point of use, no smoking zone
Res: 3 (Low)

Night Works
Hazard & Consequence: Poor visibility / fatigue
Init Risk: 15 (High)
Controls & Hold Points: Lighting survey (min lux at task), fatigue roster limits, hi-vis Class 3
Res: 5 (Low)

Heat Stress Management
Hazard & Consequence: Heat exhaustion / heat stroke
Init Risk: 16 (High)
Controls & Hold Points: WBGT monitoring, shaded rest + cool water, acclimatisation schedule, buddy checks
Res: 4 (Low)

Activity Matrix - Continuation Part 3

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Dust-Generating Works
Hazard & Consequence: Respirable crystalline silica
Init Risk: 16 (High)
Controls & Hold Points: Water suppression, on-tool extraction, FFP3 face-fit tested, health surveillance
Res: 4 (Low)

Noise-Intensive Works
Hazard & Consequence: Noise-induced hearing loss
Init Risk: 12 (Med)
Controls & Hold Points: Noise zones signed, hearing protection mandatory >85 dB(A), plant selection low-noise first
Res: 3 (Low)

Traffic Management - Internal Roads
Hazard & Consequence: Vehicle collision at pinch points
Init Risk: 16 (High)
Controls & Hold Points: One-way system, 10 km/h limit, passing bays, trained traffic marshals
Res: 4 (Low)

Welfare - Water, Rest, Sanitation
Hazard & Consequence: Dehydration / hygiene illness
Init Risk: 9 (Med)
Controls & Hold Points: Potable water points, shaded rest areas, serviced sanitation per headcount ratio
Res: 2 (Low)

Activity Matrix - Continuation Part 3

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Adverse Weather - Lightning & Storm
Hazard & Consequence: Lightning strike / crane instability
Init Risk: 15 (High)
Controls & Hold Points: 30-30 lightning rule, crane stand-down limits, secure loose materials
Res: 5 (Low)

Housekeeping - Access Routes
Hazard & Consequence: Slips, trips and falls (same level)
Init Risk: 12 (Med)
Controls & Hold Points: Designated walkways kept clear, cable covers, daily supervisor walk-round
Res: 3 (Low)

Visitor & Delivery Driver Safety
Hazard & Consequence: Unfamiliar persons in live work zones
Init Risk: 12 (Med)
Controls & Hold Points: Escorted access only, driver briefing card, PPE issue at gate
Res: 3 (Low)

Document Control & Record Retention
Hazard & Consequence: Superseded revision used in field
Init Risk: 9 (Med)
Controls & Hold Points: Controlled copy register, revision check at TBT, obsolete copies withdrawn
Res: 2 (Low)

Activity Matrix - Continuation Part 3

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Incident Reporting & Escalation
Hazard & Consequence: Delayed reporting / evidence loss
Init Risk: 12 (Med)
Controls & Hold Points: Immediate verbal + 24h written report, scene preservation, investigation per severity matrix
Res: 3 (Low)

Subcontractor Coordination (SIMOPS)
Hazard & Consequence: Conflicting simultaneous operations
Init Risk: 16 (High)
Controls & Hold Points: Daily SIMOPS meeting, shared permit board, interface risk review [HOLD POINT]
Res: 4 (Low)

Manual Handling - General
Hazard & Consequence: Musculoskeletal injury
Init Risk: 12 (Med)
Controls & Hold Points: Mechanical aids first, load weight labels, manual handling training records
Res: 3 (Low)

Storage of Flammables & Gas Cylinders
Hazard & Consequence: Fire / cylinder projectile
Init Risk: 16 (High)
Controls & Hold Points: Ventilated cage, upright chained cylinders, separation distances, signage
Res: 4 (Low)

Activity Matrix - Continuation Part 3

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Excavation Services - Trial Pits
Hazard & Consequence: Unknown service strike
Init Risk: 20 (High)
Controls & Hold Points: Drawings + CAT scan + hand-dig only within 500 mm of marked services [HOLD POINT]
Res: 5 (Low)

Concrete Washout
Hazard & Consequence: Alkaline runoff to ground/drain
Init Risk: 9 (Med)
Controls & Hold Points: Designated lined washout pit only, pH check before disposal
Res: 2 (Low)

Tower Crane - Climbing Operations
Hazard & Consequence: Structural overload during climb
Init Risk: 25 (High)
Controls & Hold Points: OEM climbing procedure, specialist crew, wind hold limits [HOLD POINT: climb permit]
Res: 5 (Low)

Demolition - Soft Strip
Hazard & Consequence: Hidden asbestos / structural instability
Init Risk: 20 (High)
Controls & Hold Points: Pre-demolition survey, licensed removal where identified, engineer sequence sign-off [HOLD POINT]
Res: 5 (Low)

Activity Matrix - Continuation Part 3

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Piling - Bored Operations
Hazard & Consequence: Auger entanglement / bore collapse
Init Risk: 20 (High)
Controls & Hold Points: Auger guards, exclusion zone, bore capping same shift
Res: 5 (Low)

Roadworks Interface - Public Highway
Hazard & Consequence: Public vehicle incursion
Init Risk: 20 (High)
Controls & Hold Points: Approved traffic management layout, advance signage, barrier-rated separation [HOLD POINT]
Res: 5 (Low)

Confined Welfare - Drinking Water Quality
Hazard & Consequence: Contaminated supply illness
Init Risk: 9 (Med)
Controls & Hold Points: Tested potable supply, dated containers, cleaning schedule
Res: 2 (Low)

Fatigue - Extended Shifts
Hazard & Consequence: Error from exhaustion on safety-critical tasks
Init Risk: 15 (High)
Controls & Hold Points: Max shift + rest rules, no safety-critical solo work after extended hours
Res: 4 (Low)

Activity Matrix - Continuation Part 3

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Young & Inexperienced Workers
Hazard & Consequence: Unrecognised hazards
Init Risk: 12 (Med)
Controls & Hold Points: Named supervisor, restricted task list, enhanced supervision ratios
Res: 3 (Low)

Emergency Drill Programme
Hazard & Consequence: Unpractised evacuation / slow muster
Init Risk: 12 (Med)
Controls & Hold Points: Monthly drill rotation (fire, rescue, spill), timed muster with roll call
Res: 3 (Low)

First Aid Provision
Hazard & Consequence: Inadequate response to injury
Init Risk: 12 (Med)
Controls & Hold Points: Trained first aiders per shift ratio, stocked stations, hospital route posted
Res: 3 (Low)

PPE Issue & Compliance
Hazard & Consequence: Non-use / damaged PPE
Init Risk: 9 (Med)
Controls & Hold Points: Free issue + fit checks, PPE matrix by task, supervisor spot checks
Res: 2 (Low)

Activity Matrix - Continuation Part 3

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Safety Signage & Barriers Audit
Hazard & Consequence: Missing or ignored warnings
Init Risk: 9 (Med)
Controls & Hold Points: Weekly audit walk, multilingual pictorial signage, barrier integrity checks
Res: 2 (Low)

Toolbox Talk Programme
Hazard & Consequence: Generic talks with no task relevance
Init Risk: 9 (Med)
Controls & Hold Points: Task-specific TBT before each shift, attendance signed, topics logged weekly
Res: 2 (Low)

Management Site Tours
Hazard & Consequence: Hazards normalised at worker level
Init Risk: 9 (Med)
Controls & Hold Points: Weekly leadership tour with close-out actions tracked to completion
Res: 2 (Low)

Change Management - Scope Variation
Hazard & Consequence: Unassessed new hazards from changed method
Init Risk: 16 (High)
Controls & Hold Points: MOC form before variation starts, RA revision re-briefed to crew [HOLD POINT]
Res: 4 (Low)

Activity Matrix - Continuation Part 3

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Close-Out - Demobilisation
Hazard & Consequence: Residual hazards left for others
Init Risk: 10 (Med)
Controls & Hold Points: Demob checklist, services capped and marked, final joint inspection
Res: 3 (Low)

Lessons Learned Register
Hazard & Consequence: Repeat incidents across packages
Init Risk: 9 (Med)
Controls & Hold Points: Register reviewed monthly, actions briefed at TBT, trends to management review
Res: 2 (Low)

Audit & Inspection Schedule
Hazard & Consequence: Drift from planned controls
Init Risk: 9 (Med)
Controls & Hold Points: Weekly HSE inspection, monthly audit, findings tracked with owners and dates
Res: 2 (Low)

Permit Office - Daily Board Control
Hazard & Consequence: Conflicting live permits
Init Risk: 16 (High)
Controls & Hold Points: Permit board reviewed at shift start, conflicts resolved before issue [HOLD POINT]
Res: 4 (Low)

Activity Matrix - Continuation Part 3

Activity-based assessment continued - initial risk scored before controls, residual risk after controls are verified in the field.

Site Access & Security
Hazard & Consequence: Unauthorised entry / public interface
Init Risk: 12 (Med)
Controls & Hold Points: Turnstile or gate log, visitor induction, hoarding integrity checks
Res: 3 (Low)

Temporary Electrics - Distribution
Hazard & Consequence: Damaged leads / RCD failure
Init Risk: 15 (High)
Controls & Hold Points: 110V or RCD-protected supplies, weekly PAT/inspection register, cable ramps
Res: 4 (Low)

Hot Works - Welding & Cutting
Hazard & Consequence: Fire / sparks to combustibles
Init Risk: 20 (High)
Controls & Hold Points: Hot work permit, fire watch during + 60 min after, extinguishers at point of work [HOLD POINT]
Res: 5 (Low)

Work at Height - Scaffold Use
Hazard & Consequence: Fall of person / incomplete scaffold
Init Risk: 25 (High)
Controls & Hold Points: Scafftag system, weekly + after-alteration inspections, no unauthorised alterations [HOLD POINT]
Res: 5 (Low)

Residual Risk Summary & ALARP Statement

All activities are reduced to Low/Medium residual risk once the stated controls are verified in place. No activity may start on initial risk alone.

High (15-25) initial
Count of Activities: Assessed per activity matrix
Action Rule: Controls + hold points mandatory before start
Authority: Site HSE authority

Medium (8-12) residual
Count of Activities: Monitor at supervision level
Action Rule: Daily TBT reinforcement + spot checks
Authority: Supervisor

Low (1-6) residual
Count of Activities: Majority of activities after controls
Action Rule: Proceed under permit and supervision
Authority: Permit issuer

ALARP Statement
Count of Activities: Further risk reduction considered for every High initial item
Action Rule: Grossly disproportionate measures documented, not silently dropped
Authority: HSE Officer

Hold Point Register

Work must stop at each hold point until the named authority verifies and releases it. Hold points are never self-released.

HP-01 Permit Verification
Where It Applies: Start of each high-risk activity
Released By: Permit issuer
Evidence: Signed permit at workface

HP-02 Isolation / LOTO
Where It Applies: Energy isolation before work
Released By: Authorised isolator
Evidence: Lock register + prove-dead test

HP-03 Pre-Start Inspection
Where It Applies: Plant, scaffold, lifting gear
Released By: Competent inspector
Evidence: Tag / inspection record

HP-04 Authority Sign-Off
Where It Applies: Document approval before execution
Released By: Client/site HSE authority
Evidence: Signed approval block

Review, Approval & Revision Record

This document is a professional draft. Field use requires review and formal approval by the client/site HSE authority.

Prepared By
Name / Organisation: SafetyWithAthar (remote)
Responsibility: Draft preparation from supplied work-pack inputs
Status: Rev. 0 issued

Reviewed By
Name / Organisation: [SITE INPUT REQUIRED]
Responsibility: Site HSE review against actual conditions
Status: [HOLD POINT]

Approved By
Name / Organisation: [SITE INPUT REQUIRED]
Responsibility: Client / site HSE authority approval
Status: [HOLD POINT]

Revision History
Name / Organisation: Rev. 0 - first issue for review
Responsibility: Superseded revisions withdrawn from field use
Status: Controlled